

BELLABEAT PLAYING IT SMART

Bellabeat, a high-tech manufacturer of health-focused products for women, and the goal is collecting data on activity, sleep, stress, and reproductive health, which has allowed Bellabeat to empower women with knowledge about their own health and habits.

Founders: Urška Sršen and Sando Mur

It was founded in 2013, and Bellabeat has grown rapidly and quickly positioned itself as a tech-driven wellness company for women.

Urška Sršen, is a cofounder and Chief Creative Officer of Bellabeat.

She anticipates that an analysis of smart device fitness data could be a gateway for opportunities for the company's growth.

Bellabeat has several products, and for this case study, I have been tasked to focus on one and analyze smart device data to gain insight into how consumers are using their smart devices. The analysis based on insights discovered will be a guide for the company's marketing strategy.

For the purpose of the analysis, I will focus on the application and give my insights on how physical activities especially total daily steps relates to the sedentary lifestyle of the users on daily basis.

The Bellabeat app feeds the users with health-related data such as their daily total steps, sleep, stress, menstrual cycle, and mindfulness habits. This data can help users understand their current practices and enables them to make healthy decisions.

This study will pass through the six phases of the data analysis which are;

- Ask
- Prepare
- Process
- Analyze
- Share
- Act

ASK

Focusing on the Bellabeat Application, the business task is to analyze data and get insights on how other smart device users are using their devices to monitor and improve their health and wellness. This will help them develop marketing strategy to unlock opportunities for the company's growth.

The key stakeholders:

- Urška Sršen - Chief Creative Officer and Bellabeat's Co-founder.
- Sando Mur - Mathematician and Bellabeat's Co-founder.
- Bellabeat's marketing analytics team - data analysis team.

PREPARE

The dataset used for this case study is found in the link below.

<https://www.kaggle.com/datasets/arashnic/fitbit/data>

Owner: Mobius (Kaggle)

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Source / Methodology: Datasets were generated by respondents to a distributed survey via Amazon Mechanical Turk between 03.12.2016–05.12.2016 and 30 eligible Fitbit users consented to the submission of personal tracker data.

<https://zenodo.org/record/53894#.X9oeh3Uzaao>

Data was downloaded in a ZIP format and had to be extracted into CSV files. The file had 18 datasets which I will be using 6, focusing on daily track records which involves total steps and sedentary activities. I went through the datasets and I observed the following:

- The data are stored in long format which makes each ID data occur in several rows.
- According to our data source, zenodo.org, 30 eligible Fitbit users consented to the submission of personal tracker data, including minute-level output for physical activity, heart rate, and sleep monitoring.

However, after viewing the files, as I preferred to use for the case study, the files `dailyActivity_merged`, `dailyCalories_merged`, `dailyIntensities_merged`, `dailysteps_merged` file shows 33 users while `sleepday_merged` shows 24 users, and `weightLogInfo_merged` shows 8 users.

Also,

Dataset was collected in 2016, which makes it a bit outdated for the analysis. The data wasn't gender specific, no age group stated and lacked location from where the sample was taken. This will create an impact on the analysis and recommendation since Bellabeat targets women.

In view of that, I concluded the data did not pass integrity, credibility.

- Moreover, while it lacks the mentioned specifics, I can say the population size used is not enough, and it makes it hard to believe the data is a true representation of its population. perhaps a larger population size could have been used in order to avoid high tendency of bias.

PROCESS

I used the following 6 datasets for the analysis based of my task and narrowed focal points:

- `dailyactivity_merged`
- `dailycalories_merged`
- `dailyIntensities_merged`
- `dailysteps_merged`
- `sleepday_merged`
- `weightLogInfo_merged`

R studio was used for data cleaning.

Steps taken for cleaning data:

1. Because the datasets had different location, First, I needed to set my working directory to the location of the CSV files using the **`setwd()`** function. Datasets were loaded using the **`read.csv()`** function.
2. Datasets were assigned to data frames, and were viewed using the **`View()`** function
3. I used the **`clean_names()`** function to ensure that the data column names are unique and consistent, that there's only characters, numbers, and underscores in the names.
4. I checked for duplicates using the **`duplicate()`** function and 3 rows were found duplicate in **`sleep_day`**. Thus **`nrows(sleep_day[duplicate(sleep_day), ])`**. They were omitted by using the function **`sleep_day <- sleep_day[!duplicated(sleep_day), ]`**
5. I sorted the data by activity **`dates`** using the **`arrange()`** function
6. Data type for date column in all tables was changed from **`char`** to **`Date`** and **`ymd`** format in order to eliminate rows that doesn't follow such format

ANALYZE

R Studio was used for the analysis.

Steps Taken:

1. I explored the data by gathering some statistics using the **head()**, **View()** and **colnames()** functions.
2. I performed data aggregation to enable me make comparisons, find trends and relationships between the 4 dataframes whose data was collected on daily basis.

The following tables **daily activity**, **daily_calories**, **daily_intensities** and **daily_steps** were compared using the **compare_df_cols()** function to find trends and relationships.

When data was aggregated, 16 column names were observed, whereas 12 column names matched among all the dataframes, they were related and similar.

However, there were 4 distinct column names that mismatched and was part of **daily_activity** dataframe.

In view of that, we will ignore the 3 similar dataframes and work with **daily_activity**, limiting us to make our analysis with 3 distinct data including **sleep_day** and **weight_log**.

Data for Analysis:

daily_activity

sleep_day

weight_log.

3. I checked how many unique participants are there in each table using **n_distinct()** function.

Results:

daily_activity = 33

sleep_day = 24

weight_log = 8

```

136 n_distinct(daily_activity$id)
137 n_distinct(sleep_day$id)
138 n_distinct(weight_log$id)
139
137:25 (Top Level)

```

```

R 4.4.2 C:/Users/MyPC/Desktop/CASE STUDY/
> n_distinct(daily_activity$id)
[1] 33
> n_distinct(sleep_day$id)
[1] 24
> n_distinct(weight_log$id)
[1] 8

```

From the above results, it is observed that **weight_log** table has the least participants and the figure will not be appropriate to be included in our analysis, hence limiting us to the remaining two.

4. I also checked the number of observations in both dataframes using the **nrows()** function.

Results:

daily_activity = 940

sleep_day = 410

```

140 nrow(daily_activity)
141 nrow(sleep_day)
142
141:16 (Top Level)

```

```

R 4.4.2 C:/Users/MyPC/Desktop/CASE S
> nrow(daily_activity)
[1] 940
> nrow(sleep_day)
[1] 410

```

For us to derail from the purpose of the analysis, I had to reconfirm the number users by their ids and the number of times these ids were logged in while they wore their Fitbit watches. Using the **count()** with **arrange()** functions, the results from our summary as below:

User ids = 33

```

150
151 summary(user_logs$logins)
150:1 (Top Level)
Console Terminal x Background Jobs x
R 4.4.2 C:/Users/MyPC/Desktop/CASE STUDY/
> view(user_logs)
> summary(user_logs$logins)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  4.00   29.00   31.00   28.48   31.00   31.00

```

From the summary above, we can say:

Min logins = 4

Max logins = 31

Average = 28

Using the **sum()** function to find the total logins confirms **940** as we know from onset.

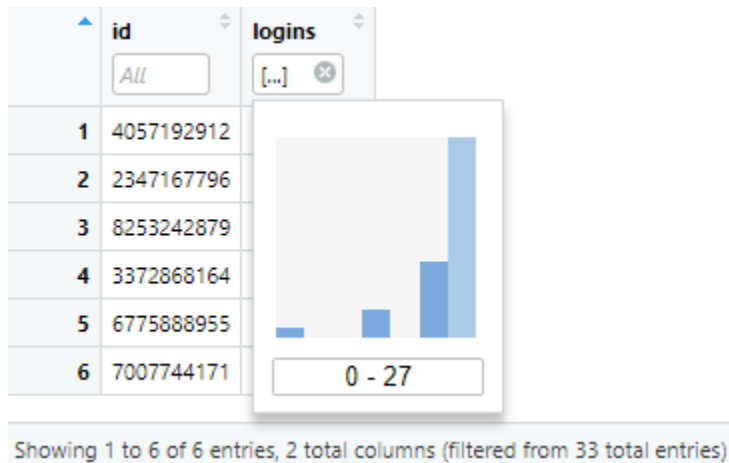
I went further to calculate how many ids logged in and how many times for the period.
Results:

logins	id_count
31	21
30	3
29	2
26	2
28	1
20	1
19	1
18	1
4	1

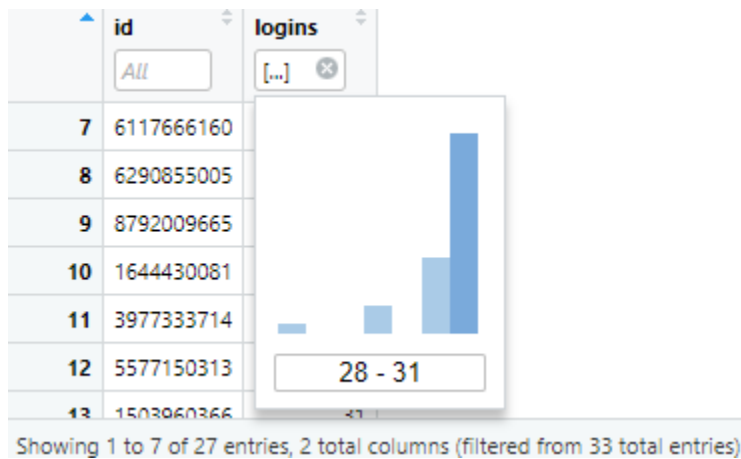
Using 28 as our average logins and 33 ids , I grouped the users into two categories based on:

users above average:

users below average:



6 ids - below average logins = (18.1 %)



27 ids – above average = 81.8 %

From the summary above, it was observed that 81.8 % of the total users are wearing the Fitbit watch and logged in. But do they keep the track records for to help them to make decisions concerning their health? We will find out.

- We will create new two data frames with the relevant data for our analysis to find trends and relationship.

From the data frames, there are specific columns that I identified to be key and to focus on for the analysis.

➤ Dataframe 1 (**new_daily_activity**):

- *Id*
- *average steps (mean_steps)* from total_steps
- *average sedentary (mean_sedentary)* from sedentary_minutes

In other for me to be realistic, I made a bit of research to know how walking can impact our health. I needed more information and statistics for a reliable source to come up with a standard while I make my analysis on the total_steps of the 33 participants.

According to the World Health Organization, <https://www.who.int/news-room/fact-sheets/detail/physical-activity>

- walking significantly impacts your body by improving cardiovascular health, strengthening bones and muscles, maintaining a healthy weight, reducing the risk of noncommunicable diseases like heart disease, stroke, type 2 diabetes, and some cancers, and positively impacting mental well-being by reducing stress and improving mood; essentially, regular walking contributes to overall health and quality of life.

According to the National Health Institute, at a glance, <https://www.nih.gov/news-events/nih-research-matters/number-steps-day-more-important-step-intensity#:~:text=Taking%204%2C000%20or%20fewer%20steps,been%20done%20in%20older%20adults.>

- Step intensity (number of steps per minute) didn't influence the risk of death, suggesting that the total number of steps per day is more important than intensity.

From the summary made by <https://www.medicalnewstoday.com/articles/how-many-steps-should-you-take-a-day#for-general-health>.

Sedentary < 5000 steps

Low active 5000 ≤ 7999 steps

Somewhat Active 8000 ≤ 9999 steps

Active > 10000 steps

More Active > 12000 steps

I will be using the above fact finding to classify the total_steps for each id

However, since each user id has multiple entries, I had to find the average total steps as **average_steps** for each user id and categorize them based on the research above, rounded to whole figures.

```

153 average_steps <- daily_activity %>%
154   select(id, total_steps) %>%
155   arrange(id) %>%
156   group_by(id) %>%
157   summarise(mean_steps = round(mean(total_steps), digits = 0))
158   view(average_steps)
159
157:63 (Top Level) ↕

```

	id	mean_steps
1	1503960366	12117
2	1624580081	5744
3	1644430081	7283
4	1844505072	2580
5	1927972279	916
6	2022484408	11371
7	2026352035	5567
8	2320127002	4717
9	2347167796	9520
10	2873212765	7556

i 23 more rows
i Use `print(n = ...)` to see more rows

The image above prints the average total steps for each user id as **mean_steps**

The above data shows the following:

- Sedentary lifestyle - **8 users**
- Low active – **11**
- Somewhat active - **7**
- Active - **4**
- More active – **3**

Note: the above sums to 33 user Ids making it accurate.

```

162 average_sedentary <- daily_activity %>%
163   select(id, sedentary_minutes) %>%
164   arrange(id) %>%
165   group_by(id) %>%
166   summarise(mean_sedentary = round(mean(sedentary_minutes), digits = 2))
167   view(average_sedentary)
168:75 (Top Level) ↕

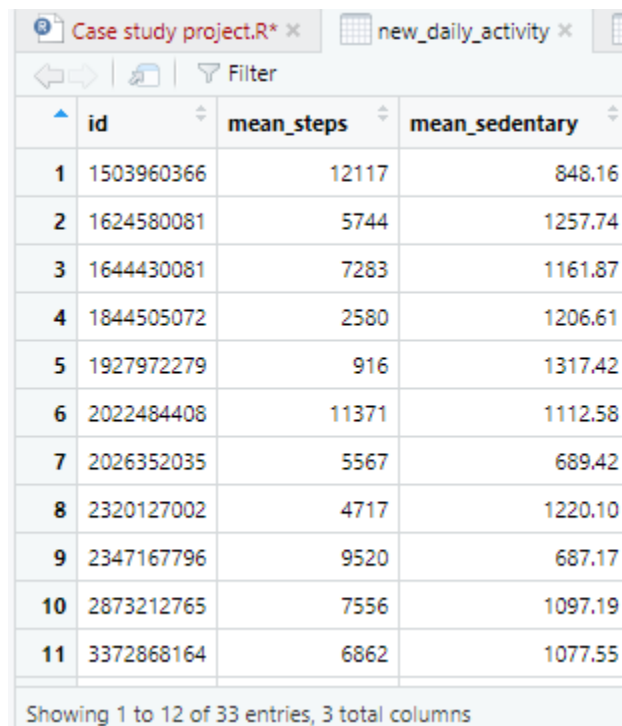
```

	id	mean_sedentary
1	1503960366	848.
2	1624580081	1258.
3	1644430081	1162.
4	1844505072	1207.
5	1927972279	1317.
6	2022484408	1113.
7	2026352035	689.
8	2320127002	1220.
9	2347167796	687.
10	2873212765	1097.

i 23 more rows
i Use `print(n = ...)` to see more rows

The image above is the print of average sedentary minutes of the user ids as **mean_sedentary**

To generate our plot to determine the relationship between the **average_steps** and **average_sedentary**, I merged the two dataframes by the two columns (*mean_steps and mean_sedentary*) as **new_activity** with total of **33 ids**.



The screenshot shows an RStudio window with a file named 'Case study project.R*' and a plot window titled 'new_daily_activity'. The main window displays a table with 3 columns: 'id', 'mean_steps', and 'mean_sedentary'. The table contains 11 rows of data, with the first 10 rows numbered 1 through 10. The status bar at the bottom indicates 'Showing 1 to 12 of 33 entries, 3 total columns'.

	id	mean_steps	mean_sedentary
1	1503960366	12117	848.16
2	1624580081	5744	1257.74
3	1644430081	7283	1161.87
4	1844505072	2580	1206.61
5	1927972279	916	1317.42
6	2022484408	11371	1112.58
7	2026352035	5567	689.42
8	2320127002	4717	1220.10
9	2347167796	9520	687.17
10	2873212765	7556	1097.19
11	3372868164	6862	1077.55

Using the syntax:

```
new_daily_activity <- merge(average_steps, average_sedentary, by = "id")
View(new_daily_activity)
```

The above image shows the print of merged average columns from our focal dataframes for plot

➤ Dataframe 2 (**new_sleep_day**)

- *Id*
- *average_min_asleep (mean_min_asleep) from total_minutes_asleep*
- *average time in bed (mean bed time) from total time in bed*

```

174 average_min_asleep <- sleep_day %>%
175   select(id, total_minutes_asleep) %>%
176   arrange(id) %>%
177   group_by(id) %>%
178   summarise(mean_min_asleep = round(mean(total_minutes_asleep), digits = 0))
179 view(average_min_asleep)

```

18:11 (Top Level) ↕

Console Terminal Background Jobs

R • R 4.4.2 • C:/Users/MyPC/Desktop/CASE STUDY/

```

id mean_min_asleep
<dbl> <dbl>
1 1503960366 360
2 1644430081 294
3 1844505072 652
4 1927972279 417
5 2026352035 506
6 2320127002 61
7 2347167796 447
8 3977333714 294
9 4020332650 349
10 4319703577 477
# i 14 more rows
# i use 'print(n = ...)' to see more rows

```

The above prints the average minutes asleep as **mean_min_asleep** from total minutes asleep.

After finding the averages of total minutes asleep and total time in bed, I merged these two columns from their dataframes, hence **new_sleep_day** with total of 24 user ids as below:

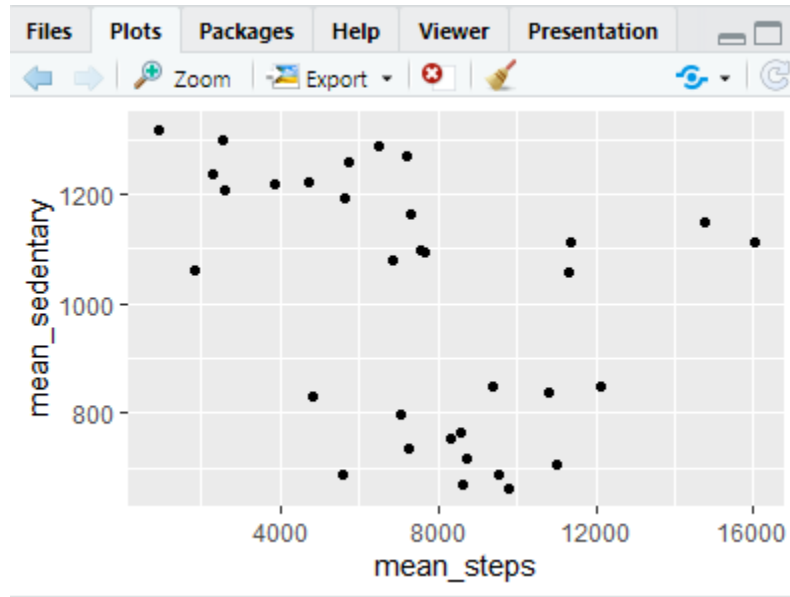
Case study project.R* × new_sleep_day × new_daily_a			
Filter			
	id	mean_bed_time	mean_min_asleep
1	1503960366	383	360
2	1644430081	346	294
3	1844505072	961	652
4	1927972279	438	417
5	2026352035	538	506
6	2320127002	69	61
7	2347167796	491	447
8	3977333714	461	294
9	4020332650	380	349
10	4319703577	502	477
11	4388161847	423	400

Showing 1 to 12 of 24 entries, 3 total columns

Based on the above, we will find the trend on how people use the app on daily basis and relate it to their health then we can share the insights to the department to enable them strategize their marketing pattern.

- We will make some plots using **ggplot()** to find the relationship between :
 - mean_steps* and *mean_sedentary*
 - mean_bed_time* and *mean_min_asleep*

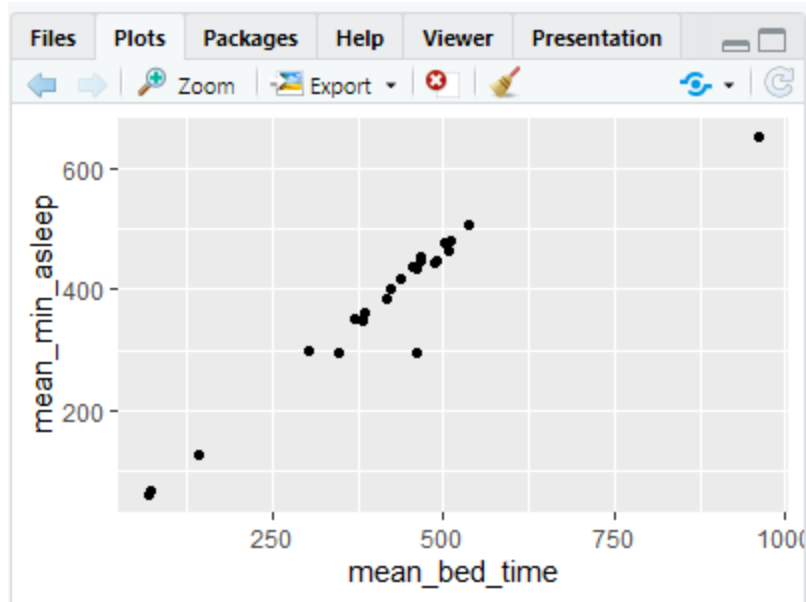
Results:



The above represents a scatter plot of data of the average steps and average sedentary of each user id. To translate the relationship between the data.

From the plot we realize that, the line plotted is not smooth, which means the mean_steps do have a meaningful relationship with the mean_sedentary. This is such that, most users who have less mean_steps below 8000 live a sedentary life, a little more between 8000 and 12000 engage in more steps than sedentary, which makes them active and finally, few are on a balanced life in way where their mean_steps balances their mean_sedentary.

Let us take a glance of the relationship between the average time in bed against average time asleep.



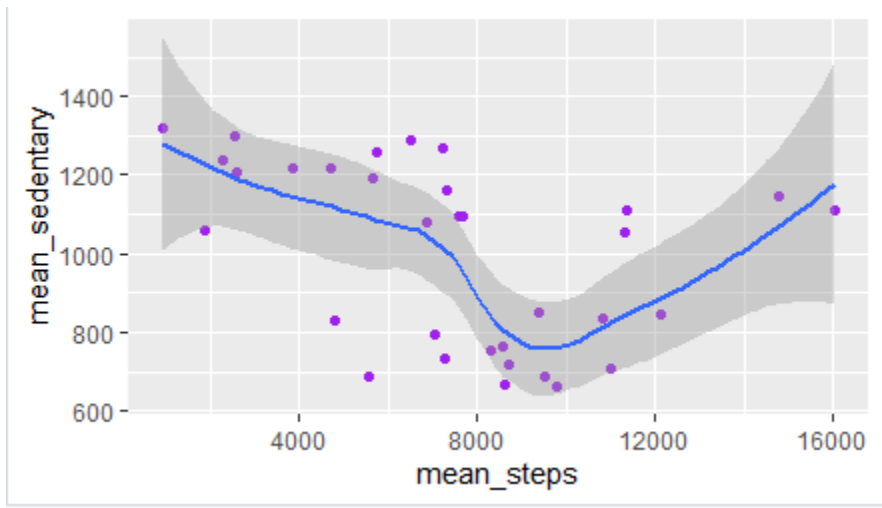
The above scatter plot shows the data of average time in bed and average minutes asleep.

There is a strong relationship between mean_bed_time and mean_min_asleep. This is such that, users do not spend more time in bed and sleep less. They balance the time spent in bed against the time asleep and that is good for their health.

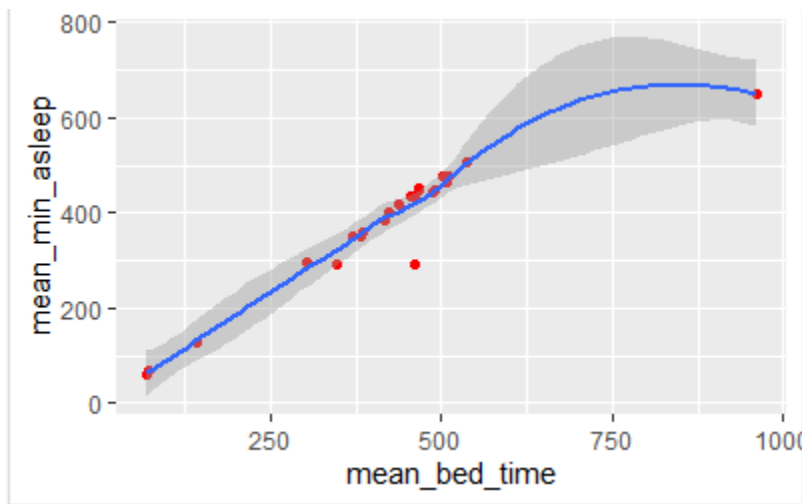
In conclusion, out of 33 participants, with 940 logins, greater number of the users are engaged more in sedentary activities and a smaller number of users are involved in activities with steps.

SHARE

- Most users about **81.8 %** of the total population are considered active users per the category because they recorded most logins, and about **18.1 %** are considered as light users. This information aided in knowing how much time is spent sedentary against any physical activity.
- There is no relationship between average total steps and average sedentary minutes as from the image below.



- 26 users out of 33, steps fall below the ideal total steps one should have thus 10000.
- 7 users were found active and met the goal of 10000 total steps on average.
- There is a strong relationship between average time spend in bed and average minutes asleep as seen from the image below.



ACT

My top 5 recommendations are as follows

- The marketing team should target users in the corporate sectors, offices and other users whose job are found to involve more sedentary.
- Users should be educated on how physical activities, exercising s such as walking, running becomes essential to their health for their well-being.
- The application should allow users set a daily/weekly/ monthly activity goal and track their progress.
- A notification feature to be added to prompt users when they are falling short of their goal. Example, a notification pop-up or continuous blink of ***“arise and move, no pain – no gain”*** as a reminder to enable them complete the activity goal.
- Although no demographic information was included for this analysis, Bellabeat is a company that caters to women, therefore the marketing team should target both women who possess trackers and those who do not. Including a period tracking option will be helpful. They will be more confident to move at that time of the month and accomplish their daily goals if they receive notifications on how to stay active during their periods.